

Containers: An Expository Account for Computer Scientists

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Abstract

This paper provides a self-contained exposition of the theory of containers, originally developed by Abbott, Altenkirch, and Ghani. We present containers as a tool for representing and reasoning about data structures in a dependently-typed setting. All proofs are given in full, using a style accessible to computer scientists who may not have extensive experience with category theory. We include numerous examples from everyday programming and discuss the relationship to polynomial functors.